1. Problem and user

Who

- Clinical officers and nurses at small clinics making triage calls under time pressure
- Ops leads who must audit decisions and control AI cost

Problem

- Paper protocols differ per site - inconsistent acuity calls, no audit trail
- Free-text notes carry phone numbers and IDs into logs
- Unclear which model/rules produced a decision, and what it cost

```
AfyaPlus: one authenticated triage service with deterministic, inspectable decisions.
```

2. Path chosen and why

Path: AfyaPlus Module 4 Capstone (recommended)

- Weeks 1-5: LLM/RAG fundamentals -> router + RAG layer
- Week 6: Triage API -> /health /token /triage with auth
- Week 7: Cost / MCP logistics -> tool stubs + cost ledger
- Week 8: Pins / versioning -> manifest + pin-field tests
- Week 9: Security -> redaction + security smoke + repo secret scan
- Week 10: Catalog -> marketplace fixture line (poster path not taken)

```
Why: one coherent product demonstrates the whole skill checklist.
Mapping: skill_matrix.md + integration_map.json (week_reuse).
```

3. Architecture - module boundaries

One ruleset feeds assessment, routing and gold labels - stub mode end to end

4. Skill coverage - which skills, where in repo

Weeks 1-5	src/afyaplus/rag/, router/deterministic.py, evaluation.py
Week 6	triage/api.py, triage/auth.py, tests/test_api.py
Week 7	mcp/tools.py, mcp/cost_tracker.py, docs/cost_analysis.md
Week 8	pins/manifest.py + manager.py, tests/test_pin_fields.py
Week 9	security/redaction.py, tests/test_security_smoke.py
Week 10	catalog/service.py, fixtures/catalog_listings_sample.jsonl
Cross	pytest.ini tiers, perf_stub.json, docs pack, git flow

```
Full mapping: skill_matrix.md (required/optional per row)
Machine readable: integration_map.json -> week_reuse
```

5. Live demo

Three commands

```
$ ./start_service.sh
$ curl -i http://127.0.0.1:8000/health
$ TOKEN=$(curl -s -X POST .../token ...) && curl -i .../triage -H "Bearer $TOKEN" ...
```

- On screen: 200 health -> token -> triage with a PII note
- Response shows redacted_note: 'Call [PHONE] or [EMAIL] about the fever'
- Server log prints the redacted note only

Backup if the network misbehaves

- docs/demo_transcript.md - 60-second step-by-step with real output
- docs/evidence/api_transcript.txt - full headers for 200/401/403/422

6. Testing evidence

```
$ pytest -q -> 55 passed, 12 deselected
$ pytest -m integration -> 12 passed, 55 deselected
$ python scripts/eval_score.py -> EVAL SCORE: 1.0000 (300/300 rows)
```

- Security smoke: real redaction test, repo secret scan, token tamper checks
- Pin test: source digest + requirements must match the manifest
- Perf (p95): route 0.05ms | twin 0.15ms | retriever 0.87ms | HTTP 18.3ms

The test that earned its keep

- Golden set caught label drift: pre-fix eval 0.6833, 95/300 rows wrong
- Fix: shared policy module - incident note in docs/incidents/

7. Security and compliance choices

- Auth: HMAC-SHA256 bearer tokens - 401 without/invalid, 403 for wrong role
- Redaction at the edge: raw PII never echoed, never logged (shown live)
- Repo hygiene: CI scans every text file for key patterns; no .env committed
- Fixtures: synthetic only, identifier fields rejected by tests
- Data lines separated: clinical fixture vs catalog fixture, asserted both ways
- Secrets from env (AFYA_TOKEN_SECRET), dev placeholder clearly labelled

```
No production data, no scraping, no live provider calls in this build.
```

8. Cost table with assumptions

Step	API calls	Cost
Unit + security + golden tests	0	$0.00
Integration tests (in-process)	0	$0.00
Golden-set eval, 300 rows	0	$0.00
Perf measurement (~1,400 iterations)	0	$0.00
Local demo (loopback HTTP)	0	$0.00
Live demo allowance (declared)	~10	~$1.00 - not spent
TOTAL SPENT	**0**	**$0.00**

- Assumptions: stub mode, no keys used, pay-per-token pricing if wired
- Teaching figures if a model were wired: small chat ~$0.001 per demo run

9. What works vs deferred

Works (evidence linked)

- One-command pipeline + eval 1.0000
- API: auth, validation, redaction
- 55 + 12 tests green (captured)
- Pins verified, perf measured
- Fixtures 300 + 200, no PII
- Incident note: drift 0.6833 fixed

Deferred (declared)

- Live LLM backend - stub path taken
- Real MCP server - fallback reason shown
- Docker build - not installed here
- Cloud deploy - no creds, docs only
- Embeddings - deterministic overlap
- Streaming / persistence - out of scope

- Eval caveat: labels share the runtime policy - regression guard, not clinical validation

10. Process

- Git: main = scaffold, feature/module4-capstone = work, merged --no-ff
- History includes the red -> green fix (06ec11a), not a squashed snapshot
- Peer review: Alvin Swalah, 12-item checklist, 5 findings - all re-verified
- CI: .github/workflows/ci.yml runs tests + eval (EVAL SCORE in logs)

Next steps

- Push the repo, open the PR, present Week 12 Monday
- Optionally wire a live model behind the same interface (budget ~$1)

```
Questions? - deck, transcript and evidence all live in this repo.
```